MILITARY SPECIFICATION

LINES, RADIO FREQUENCY TRANSMISSION

(COAXIAL, AIR DIELECTRIC), 50 OHM

All interested Commands of the Department of the Navy have concurred in the use of this specification.

1. SCOPE

1.1 This specification covers the detail requirements for 50-ohm, 1-5/8 inch aluminum, coaxial, air-dielectric transmission lines.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
WW-T-700/1 - Tube, Aluminum, Drawn, Seamless, 1100

MILITARY
MIL-L-3890 - Lines, Radio Frequency Transmission (Coaxial, Air Dielectric)
MIL-C-5541 - Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloys

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with MIL-L-3890, and as specified herein.

3.2 Material. The outer conductors of the lines shall be fabricated of aluminum 1100, temper H18 in accordance with WW-T-700/1 and shall be chemically treated inside and outside with class 3 chemical treatment in accordance with MIL-C-5541.

3.3 Design, construction, and physical dimensions. The lines shall be of the design, construction, and physical dimensions specified on figure 1.

3.4 Electrical characteristics. Electrical characteristics shall be as specified in table I.

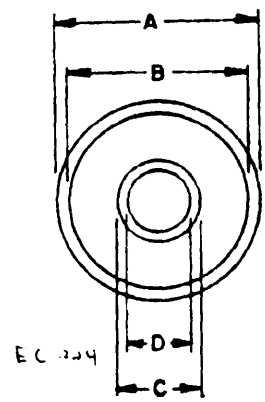
TABLE I. Electrical characteristics

Type	Line size	Operating frequency range, incl. (MHz)1/	Characteristic impedance2/ (ohms)			Test frequencies (MHz)	VSWR3/ over operating frequency band	High potential test voltage (kv)	Attenuation (db/100 ft)
			Nom	Min	Max				
RG-382/U	1-5/8	0.5 to 2,700	50	49.64	50.31	2,000	1.10:1	10.0	1.00

1/ The operating frequency range is for information only.
2/ The characteristic impedance for each line size, as shown, is exclusive of supports, and is for informational purposes.
3/ VSWR tests shall be conducted on 20-foot lengths, minimum.

FSC 5985

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Type	Line size	Outer conductor				Inner conductor			
		Diameter		Wall thickness[1]		Diameter		Wall thickness[1]	
		A	B	Nom	Max deviation from average[2]	C	D	Nom	Max deviation from average[2]
RG-382/U	1-5/8	1.625 ±0.003	1.527 ±0.003	0.049	0.005	0.6640 ±0.0025	0.5880 ±0.0025	0.038	0.005

1/ Average wall thickness is one-half the difference between the corresponding inside and outside diameters at any cross section perpendicular to the axis, measured arbitrarily at any point on the periphery.

2/ Maximum deviation of the wall thickness from the average wall thickness (eccentricity) at any cross section is the difference between the average wall thickness and either the maximum or minimum wall thickness, whichever is greater.

NOTE: Dimensions are in inches.

Figure 1. 50-ohm lines.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection shall be in accordance with MIL-L-3890.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-L-3890.

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6. NOTES

6.1 Notes. In addition to the notes specified herein, the notes specified in MIL-L-3890 are applicable to this specification.

6.2 Attenuation. The nominal attenuation of 50-ohm lines shall be as shown in table II.

TABLE II. Nominal attenuation

Type	Nominal attenuation (db/100 feet)					
	10 MHz	100 MHz	500 MHz	1,000 MHz	2,000 MHz	4,000 MHz
RG-382/U	0.058	0.20	0.48	0.68	1.00	....

6.3 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

Review activity:
Navy - SH

Preparing activity:
Navy - EC
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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